Advanced Topics in Computer Graphics II
Geometry Processing

Applications of the Laplace-Beltrami Operator

December 12, 2024

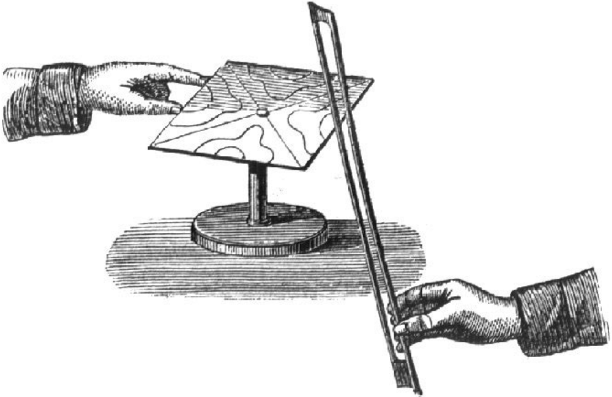
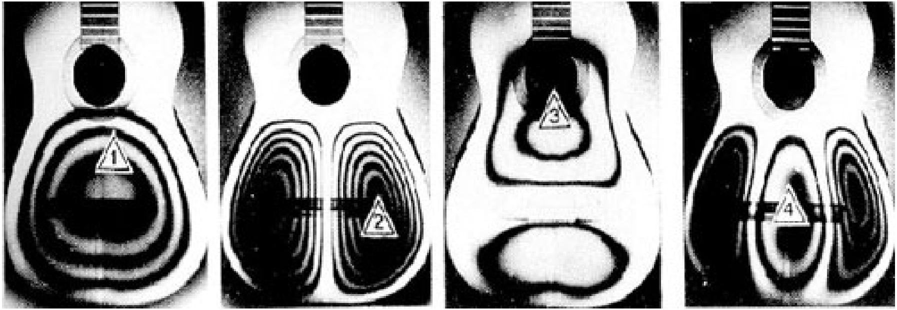


Figure: **Chladni's experimental setup allowing to visualize acoustic waves**
(Entdeckungen über die Theorie des Klanges)

Figure: Ernst Chladni (1715-1782)

Chladni plates



Patterns seen by Chladni are solutions to the **stationary Helmholtz equation**

$$\Delta_X f = \lambda f$$

Solutions of this equation are **eigenfunctions** of Laplace-Beltrami operator

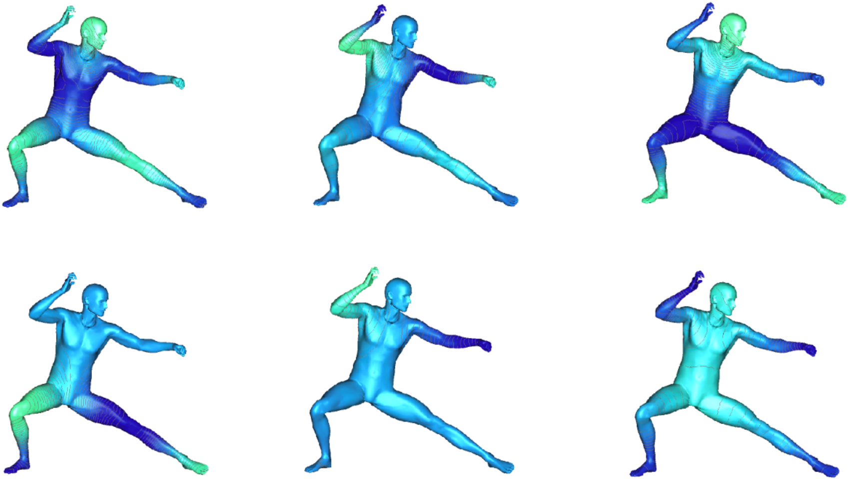


Figure: The first eigenfunctions of the Laplace-Beltrami operator

- Eigendecomposition of Laplace-Beltrami operator of compact shape gives a discrete set of eigenvalues and eigenfunctions $\{\phi_i, \lambda_i\}_{i \geq 1}$
- Since the Laplace-Beltrami operator is symmetric, eigenfunctions $\{\phi_i\}_{i \geq 1}$ form an **orthogonal basis** for $L_2(X)$
- The eigenvalues and eigenfunctions are **isometry invariant**

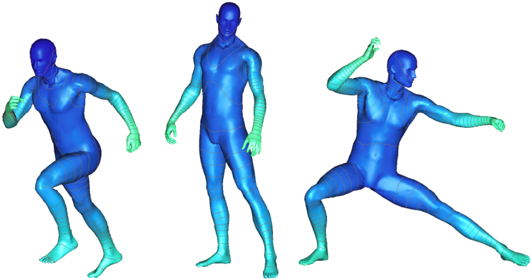
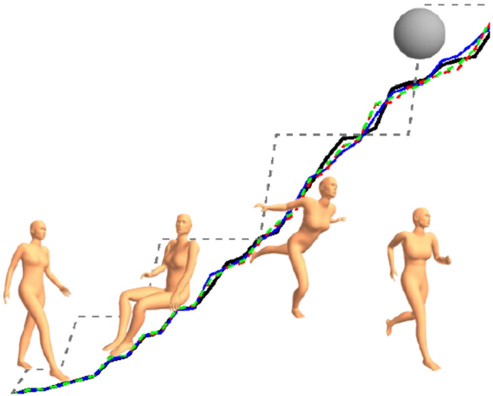


Figure: An eigenfunction of the Laplace-Beltrami operator computed on different deformations of the shape, showing the invariance of the Laplace-Beltrami operator to isometries[1]

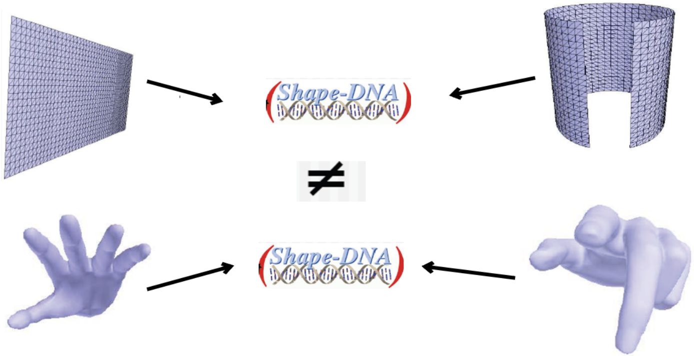
[1] M. Reuter et al. "Laplace-spectra as fingerprints for shape matching". In: *Proceedings of the 2005 ACM symposium on Solid and physical modeling*. 2005, pp. 101–106.

Use the Laplace-Beltrami spectrum as an isometry-invariant shape descriptor:
Shape DNA (Reuter et al., 2006)



Main idea (Theorem):

- Isometric Shapes $\Rightarrow$ Same spectra



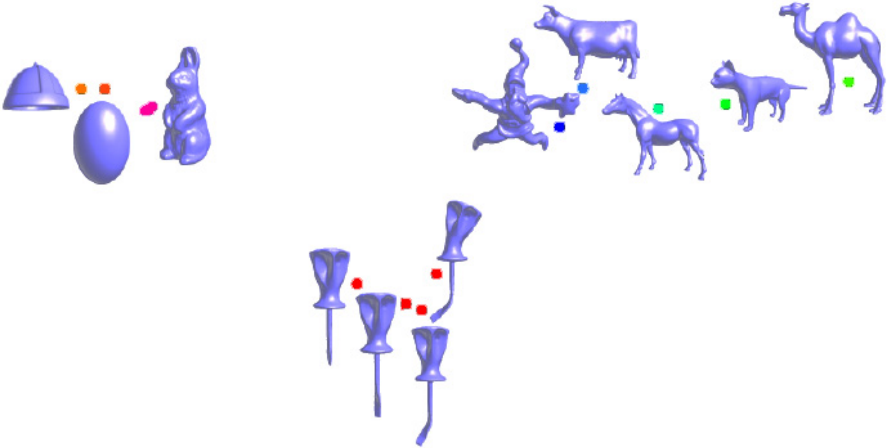


Figure: Shape similarity using Laplace-Beltrami spectrum

To hear the shape of the drum

Use the Laplace-Beltrami Spectrum $\{\lambda_i\}_{i\geq 1}$ as an isometry-invariant shape descriptor: **Shape DNA** (Reuter et al., 2006)

Is this description unique?

- Isometric shapes are isospectral (see theorem above)
- Are isospectral shapes isometric?

In Chladni's experiments, the spectrum describes acoustic characteristics of the plates ("modes" of vibrations)
What can be "heard" from the spectrum:

- Total Gaussian curvature
- Euler characteristic
- Area

"Can one hear the shape of the drum?"

Figure: Mark Kac (1914-1984)

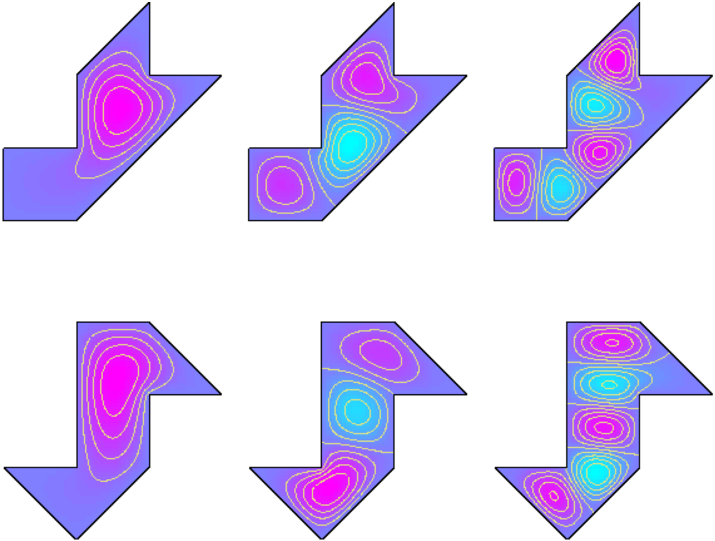


Figure: Counter-example of isospectral but not isometric shapes

The **eigenfunctions and the eigenvalues** of the Laplace-Beltrami operator **uniquely determine** the metric tensor of the shape!

- One can recover the shape up to an isometry from $\{\phi_i, \lambda_i\}_{i\geq 1}$ where ϕ_i is an eigenvector of the **Laplace-Beltrami** operator (Rustamov et al., 2007)

Definition (Global Point Signature(GPS))

For every point $\boldsymbol{p} \in \mathcal{M}$ define **G**lobal **P**oint **S**ignature

$$\mathrm{GPS}(\boldsymbol{p}) = \left(\frac{1}{\sqrt{\lambda_1}}\phi_1(\boldsymbol{p}), \frac{1}{\sqrt{\lambda_2}}\phi_2(\boldsymbol{p}), \frac{1}{\sqrt{\lambda_3}}\phi_3(\boldsymbol{p}), \dots \right)$$

GPS is a mapping of the surface onto an infinite dimensional space. Each point gets a signature.

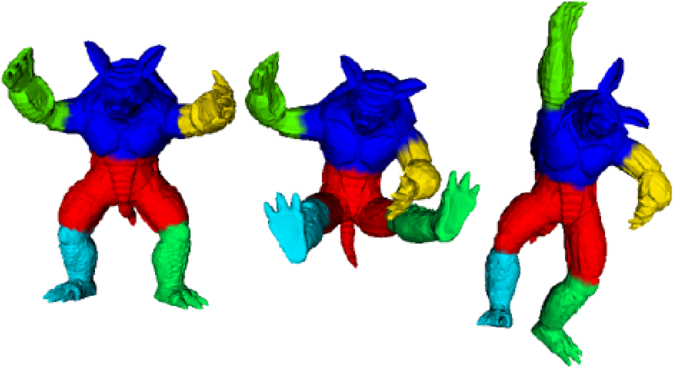
Evaluate $\phi_i(\boldsymbol{p})$ on surface via barycentric interpolation

$$\phi_i(\boldsymbol{p}) = \sum_{k=1}^{3} b_k(\boldsymbol{p})\phi_i(\boldsymbol{v}_{\boldsymbol{j_k}})$$

where $j = 1, \dots, n$ are the vertex indices and $b_k(\boldsymbol{p}), k = 1, 2, 3$ are the barycentric coordinates of the triangle containing $\boldsymbol{p}$

Properties of **GPS**

- If $\boldsymbol{p} \neq \boldsymbol{q}$, then $\mathrm{GPS}(\boldsymbol{p}) \neq \mathrm{GPS}(\boldsymbol{q})$.
- GPS is isometry invariant (since Laplace-Beltrami is)
- Given all eigenfunctions and eigenvalues, can recover the shape up to isometry (not true if only eigenvalues are known).
- K-means done on the embedding provides a segmentaion.



Properties of **GPS**

- Euclidean distances in the GPS embedding are meaningful

Need Green's function to understand this fact:
Let L be a differential operator (e.g. the Laplace-Beltrami Operator). The Green function G at a point s is the solution to following differential equation:

$$LG(x, s) = \delta(x - s)$$

Starting with an equation $Lu(x) = f$ and integrating the product of the previous function with f we get

$$\int LG(x, s) f(s) \mathrm{d}s = \int \delta(x - s) f(s) \mathrm{d}s = f(x) = Lu(x)$$

Because of the linearity of L we get the following relation between the solution of the differential equation $u(x)$ and the Green´s function:

$$\int G(x, s) f(s) \mathrm{d}s = u(x).$$

Properties of **GPS**

- Euclidean distances in the GPS embedding are meaningful

$$\int G(x,s) f(s) \mathrm{d}s = u(x)$$

Interpretation: $G(x,s)$ measures the influence of s on the solution $u(x)$.

Theorem

Let $\{\phi_i, \lambda_i\}_{i \geq 1}$ *be the eigendecomposition of* L *then*

$$G(x,y) = \sum_{i=1}^{\infty} \frac{\phi_i^*(x)\phi_i(y)}{\lambda_i}$$

Furthermore,

$$G(x,y) = \mathrm{GPS}(x) \cdot \mathrm{GPS}(y)$$

- Let
$$G(x,y) = \sum_{i=1}^{\infty} \frac{\phi_i^*(x)\phi_i(y)}{\lambda_i},$$
where $\{\phi_i, \lambda_i\}$ are eigenfunctions and eigenvalues of L.
- Applying L to $G(x,y)$:
$$\begin{aligned} LG(x,y) &= L \sum_{i=1}^{\infty} \frac{\phi_i^*(x)\phi_i(y)}{\lambda_i} \\ &= \sum_{i=1}^{\infty} \phi_i^*(x)\phi_i(y) \\ &= \delta(x-y) \end{aligned}$$
where the last equality is due to the completeness relation.
- Therefore, $LG(x,y) = \delta(x-y)$, proving that $G(x,y)$ is the Green's function for L.

- Assume the eigenfunctions $\{\phi_i(x)\}$ are orthonormal:

$$\int \phi_i^*(x)\phi_j(x)\,dx = \delta_{ij}$$

- Expressing $\delta(x-y)$ using eigenfunctions:

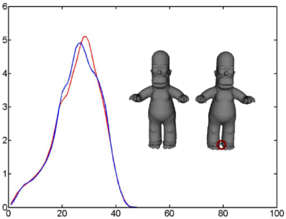
$$\begin{aligned} f(y) &= \int \delta(x-y)f(x)\,dx \\ &= \int \delta(x-y)\sum_{i=1}^{\infty} c_i\phi_i(x)\,dx = \sum_{i=1}^{\infty} c_i\phi_i(y) \\ &= \sum_{i=1}^{\infty}\left(\int f(x)\phi_i^*(x)\,dx\right)\phi_i(y) \end{aligned}$$

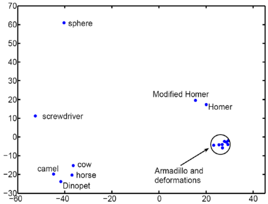
- Hence, we obtain the completeness relation:

$$\sum_{i=1}^{\infty}\phi_i^*(x)\phi_i(y) = \delta(x-y)$$

Comparing **GPS**:

- Given a shape, determine its GPS embedding.
- Construct a histogram of pairwise GPS distances [D2-distance] (note that GPS is defined up to sign flips, distances are preserved)
- For any 2 shapes, compute the L_2-norm difference between their histograms.
- For refined comparisons, use more than one histogram.





Conclusions

- Laplacian embedding is useful because of its isometry-invariance. Can be used for comparing non-rigid shapes under isometric deformations
- Sign flipping and repeated eigenvalues can cause difficulties (no canonical way to choose them).

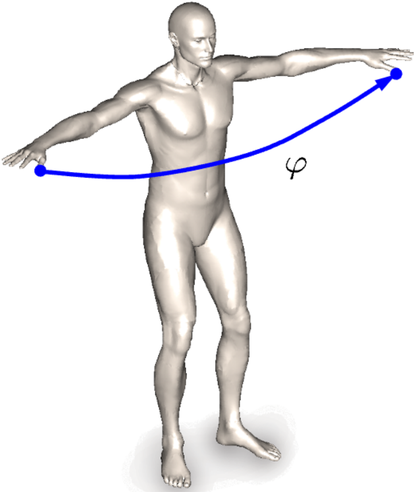
Limitations

- Embeddings are not necessarily stable or mesh independent.
- Difficult to compute for large meshes (millions of points)
- Both topological and geometric stability is not well understood.

- Shape X is **symmetric**, if there exists a **rigid motion** $f \in \mathrm{Iso}(\mathbb{R}^3)$ such that $f(X) = X$.
- Alternatively: Shape X is symmetric if there exists an automorphism $\varphi : X \to X$ such that

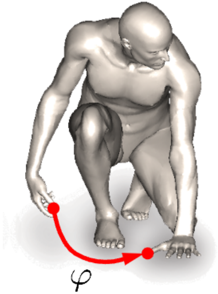
$$d_{\mathbb{R}^3}(\boldsymbol{x}, \boldsymbol{x}') = d_{\mathbb{R}^3}(\varphi(\boldsymbol{x}), \varphi(\boldsymbol{x}'))$$

- *Said differently:* Shape X is symmetric if $(X, d_{\mathbb{R}^3}|_X)$ has a non-trivial **self-isometry**
- Substitute **extrinsic** metric $d_{\mathbb{R}^3}|_X$ with **intrinsic** counterpart d_X
- Distinguish between **extrinsic** and **intrinsic** symmetry.



Extrinsic symmetry

$$d_{\mathbb{R}^3} = d_{\mathbb{R}^3} \circ (\varphi \times \varphi)$$

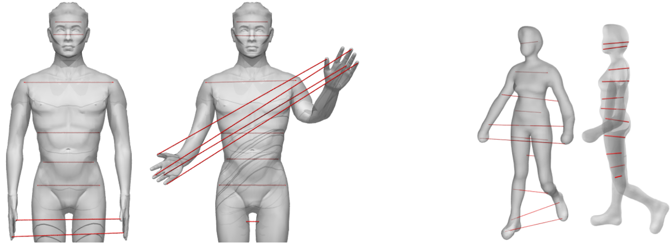


Intrinsic symmetry

$$d_X = d_X \circ (\varphi \times \varphi)$$

Problem:

- Given a shape, detect and quantify *intrinsic symmetries*



- Detect multiple symmetries efficiently. Avoid doing a non-linear optimization. Find efficient encoding for symmetries.

M. Ovsjanikov, J. Sun, et al. "Global intrinsic symmetries of shapes". In: *Computer graphics forum*. Vol. 27. 5. Wiley Online Library. 2008, pp. 1341–1348

Main Observation:

- Eigenvalues and eigenvectors of the Laplace-Beltrami operator are invariant with isometric deformations:
  If ϕ_X is an eigenfunction of Δ_X, and $T : X \to Y$ is an isometry:

$$\Delta_Y(\phi_X \circ T) = \lambda(\phi_X \circ T)$$

- If $X = Y$ and T is a self-isometry (symmetry). Then:

$$\Delta(\phi \circ T) = \lambda(\phi \circ T)$$

- Recall that non-repeating eigenfunctions are defined up to a sign, and repeating eigenfunctions span a linear subspace.

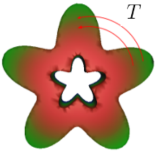
Theorem

Let O be a compact manifold having an intrinsic symmetry T, then $\mathrm{GPS}(O)$ has a Euclidean symmetry:

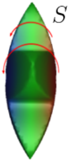
$$\left\|\mathrm{GPS}(\boldsymbol{p}_1) - \mathrm{GPS}(\boldsymbol{p}_2)\right\|_2 = \left\|\mathrm{GPS}(T(\boldsymbol{p}_1)) - \mathrm{GPS}(T(\boldsymbol{p}_2))\right\|_2, \ \forall \boldsymbol{p}_1, \boldsymbol{p}_2 \in O$$

Note: For eigenfunctions, associated with non-repeating eigenvalues there are only 2 possibilities:

Positive



or



Negative

$$\phi_i \circ T = \phi_i$$

$$\phi_i \circ T = -\phi_i$$

- Define signature, i.e., $s(\boldsymbol{x}) := \mathrm{GPS}(\boldsymbol{x})$
- Given a closed manifold $\mathcal{M}$, suppose it has an intrinsic symmetry $T : \mathcal{M} \to \mathcal{M}$
- Let $\boldsymbol{S} = (S_1, \dots, S_d)^\top$ be a sign sequence corresponding to T. For any point $\boldsymbol{x} \in \mathcal{M}$ we have:

$$S(T(\boldsymbol{x})) = (S_1 s_1(\boldsymbol{x}), S_2 s_2(\boldsymbol{x}), \dots, S_d s_d(\boldsymbol{x}))^\top$$

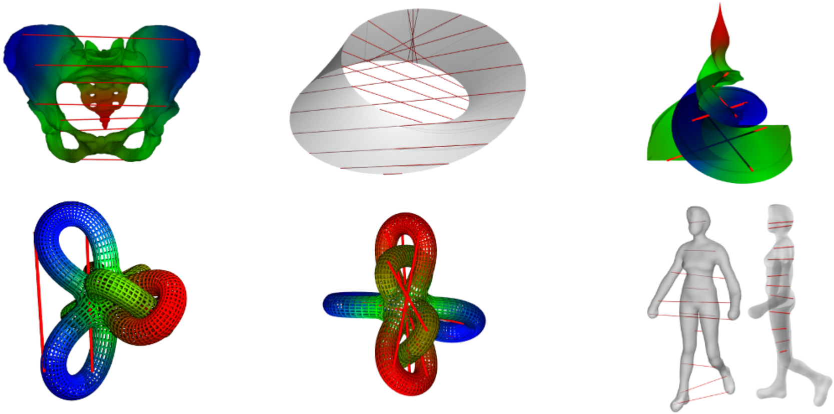
  Note, in particular, $|S(T(\boldsymbol{x})_i)| = |s(\boldsymbol{x}_i)|$, $\forall i$

- Given a sign sequence S, we can test whether this sequence is associated with any intrinsic symmetry on the manifold by modifying the signature s of each point $\boldsymbol{x}$ to $s'(\boldsymbol{x}) = (S_1 s_1(\boldsymbol{x}), S_2 s_2(\boldsymbol{x}), \dots, S_d s_d(\boldsymbol{x}))^\top$ and consider

$$E(S) = \sum_{\boldsymbol{x} \in \mathcal{M}} \min_{\boldsymbol{y} \in \mathcal{M}} \left\| s'(\boldsymbol{x}) - s(\boldsymbol{y}) \right\|_2^2$$

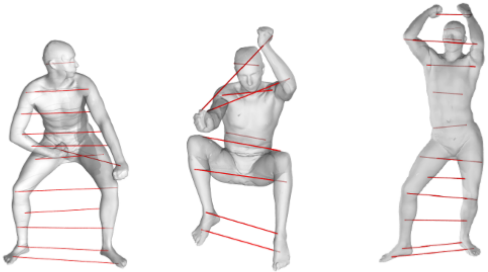
- For a sign sequence S corresponding to an intrinsic symmetry, $E(S) = 0$ because for every point there must exist a corresponding symmetric point.
- In the case of approximate symmetry, $E(S)$ must be small.
- Straight forward evaluation requires to test 2^d possibilities which is expensive
- Efficient solution is given in the paper (Ovsjanikov, Sun, et al., 2008).

Euclidean symmetries when present.
Two different symmetries for human shape.

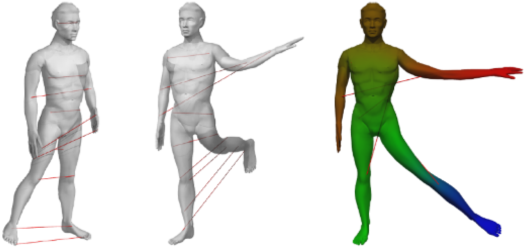


Results

SCAPE:



As-Rigid-as-possible:

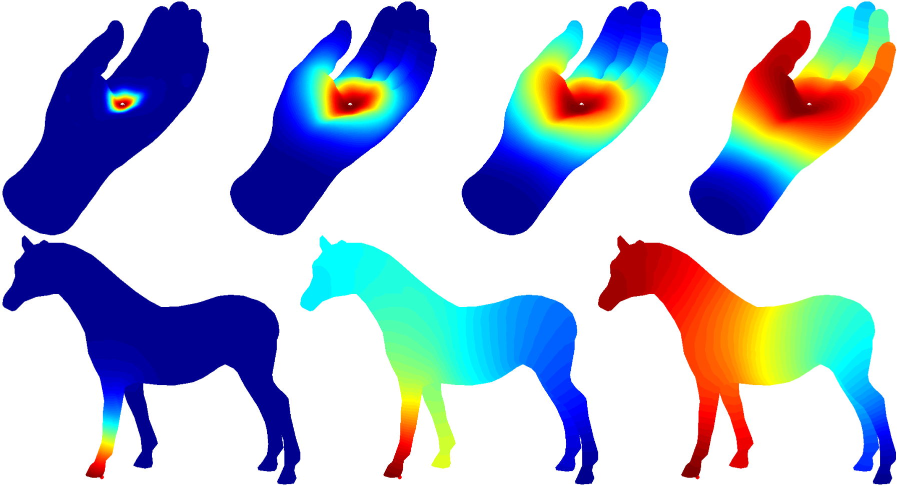


- Simple algorithm for detecting intrinsic symmetries
- Intrinsic symmetries become Euclidean in GPS
- Laplace-Beltrami Operator encodes metric properties

Limitations:

- Incorporate Repeating eigenvalues
- Disambiguate between mixed symmetries
- *Local and partial* intrinsic symmetries

Consider heat conduction on the surface



Let

- $\mathcal{M}$: compact Riemanian 2-manifold in $\mathbb{R}^3$.
- $u(x,t)$: Temperature at point $\boldsymbol{x}$ at time t
- $\Delta_{\mathcal{M}}$: Laplace-Beltrami Operator of $\mathcal{M}$
- Heat equation

$$\Delta_{\mathcal{M}} u(\boldsymbol{x},t) = -\frac{\partial u(\boldsymbol{x},t)}{\partial t}$$

- If $\mathcal{M}$ has boundaries, we require u to satisfy

$$u(\boldsymbol{x},t) = 0 \, \forall \boldsymbol{x} \in \partial\mathcal{M}, t \in \mathbb{R}$$

(Dirichlet Boundary condition)

Heat equation:

$$\Delta_{\mathcal{M}} u(\boldsymbol{x}, t) = -\frac{\partial u(\boldsymbol{x}, t)}{\partial t}, \;\; u(\boldsymbol{x}, 0) = f(\boldsymbol{x}), \;\; \boldsymbol{x} \in \mathbb{R}$$

Decomposing u into eigenfunctions of $\Delta_{\mathcal{M}}$ leads to

$$\begin{aligned} u(\boldsymbol{x}, t) &= \sum_{i=1}^{\infty} \langle u(\cdot, t), \phi_i(\cdot) \rangle \phi_i(\boldsymbol{x}) \\ \frac{\partial}{\partial t} u(\boldsymbol{x}, t) &= \sum_{i=1}^{\infty} \frac{\partial}{\partial t} \langle u(\cdot, t), \phi_i(\cdot) \rangle \phi_i(\boldsymbol{x}) \\ \Delta_{\mathcal{M}} u(\boldsymbol{x}, t) &= \sum_{i=1}^{\infty} \langle u(\cdot, t), \phi_i(\cdot) \rangle \Delta_{\mathcal{M}} \phi_i(\boldsymbol{x}) \\ &= \sum_{i=1}^{\infty} \langle u(\cdot, t), \phi_i(\cdot) \rangle \lambda_i \phi_i(\boldsymbol{x}) \end{aligned}$$

Comparing coefficients we get

$$\frac{\partial}{\partial t}\langle u(\cdot,t),\phi_i(\cdot)\rangle = -\langle u(\cdot,t),\phi_i(\cdot)\rangle\lambda_i$$

which results in solutions for the coefficients

$$\langle u(\cdot,t),\phi_i(\cdot)\rangle = d_i \mathrm{e}^{-\lambda_i t}$$

and the closed form solution for the function u

$$u(\boldsymbol{x},t) = \sum_{i=1}^{\infty} d_i \mathrm{e}^{-\lambda_i t}\phi_i(\boldsymbol{x})$$

The unknown d_i are determined by the initial value of $u(\boldsymbol{x},0)$

$$\lim_{t\to 0} u(\boldsymbol{x},t) = \lim_{t\to 0}\sum_{i=1}^{\infty} d_i \mathrm{e}^{-\lambda_i t}\phi_i(\boldsymbol{x}) = \sum_{i=1}^{\infty} d_i\phi_i(\boldsymbol{x}) = u(\boldsymbol{x},0) = f(\boldsymbol{x})$$

$$d_i = \langle f,\phi_i(\cdot)\rangle$$

Coefficients:

$$d_i = \langle f, \phi_i(\cdot) \rangle$$

$$\sum_{i=1}^{\infty} \underbrace{\left(\int_{\mathcal{M}} \phi_i(\boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{y} \right)}_{d_i} \mathrm{e}^{-\lambda_i t} \phi_i(\boldsymbol{x}) = \int_{\mathcal{M}} \sum_{i=1}^{\infty} \mathrm{e}^{-\lambda_i t} \phi_i(\boldsymbol{x}) \phi_i(\boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{y}$$

$$= \int_{\mathcal{M}} k_t(\boldsymbol{x}, \boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{y}$$

Definition

The heat kernel

$$k_t(\boldsymbol{x}, \boldsymbol{y}) := \sum_{i=1}^{\infty} \mathrm{e}^{-\lambda_i t} \phi_i(\boldsymbol{x}) \phi_i(\boldsymbol{y})$$

gives the amount of heat transferred from $\boldsymbol{x}$ to time $\boldsymbol{y}$ in time t.

Remark:

$u(\boldsymbol{x}, t)$ can be interpreted as the result of a convolution between f and the heat kernel $k_t(\boldsymbol{x}, \boldsymbol{y})$

- Physical interpretation: Let $H_t(f)$ denote the heat distribution at time t

- Solving the Heat equation leads to

$$H_t(f) = u(\boldsymbol{x}, t) = \int_{\mathcal{M}} k_t(\boldsymbol{x}, \boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{y}$$

Solution of Heat equation

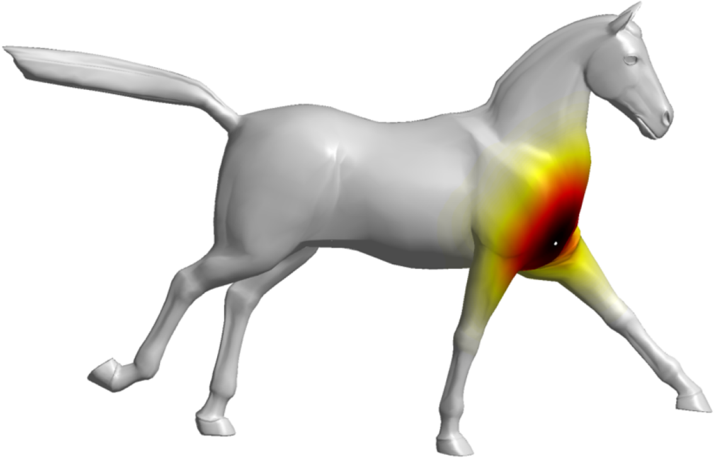
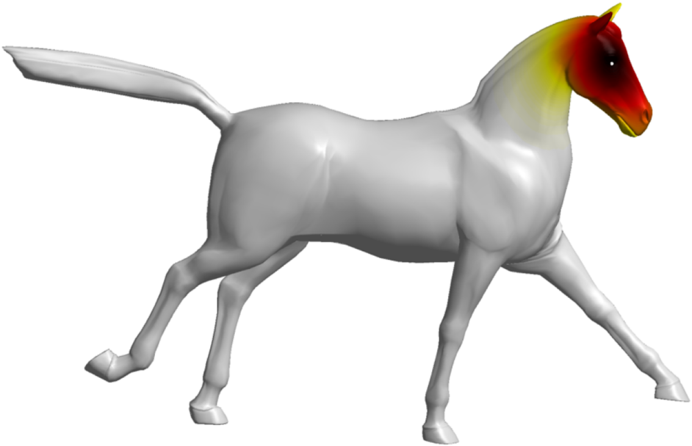


Figure: From: M. M. Bronstein et al. "Geometric deep learning: going beyond euclidean data". In: *IEEE Signal Processing Magazine* 34.4 (2017), pp. 18–42



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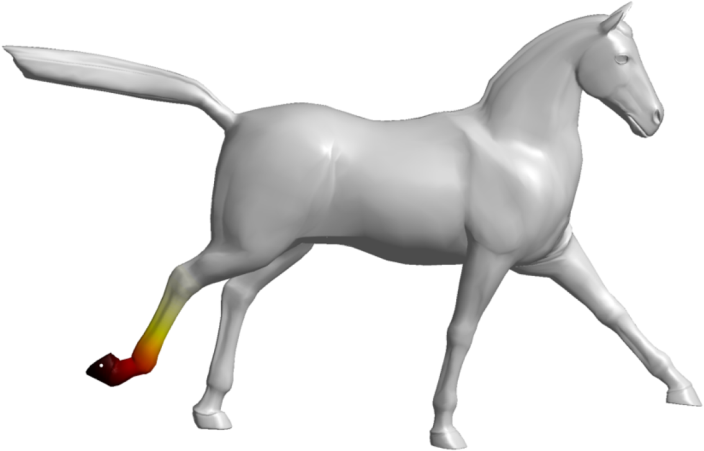


Figure: From: M. M. Bronstein et al. "Geometric deep learning: going beyond euclidean data". In: *IEEE Signal Processing Magazine* 34.4 (2017), pp. 18–42

Summary: Given

$$f : \mathcal{M} \to \mathbb{R}$$
$$H_t(f) \text{ heat distribution at time } t$$
$$\lim_{t \to 0} H_t(f) = f$$

then H_t is called the heat operator.

- Note that both, $\Delta_{\mathcal{M}}$ and H_t are operators that map one realvalued function on $\mathcal{M}$ to another such function.
- One verifies easily that
$$H_t = \mathrm{e}^{-t\Delta_{\mathcal{M}}}$$
- Thus, both operators share the same eigenfunctions and eigenvalues:

λ eigenvalue of $\Delta_{\mathcal{M}} \Leftrightarrow \mathrm{e}^{-\lambda t}$ **is eigenvalue of** H_t to the same eigenfunction.

Properties of heat kernel $k_t(\boldsymbol{x}, \boldsymbol{y})$:

- Symmetric: $k_t(\boldsymbol{x}, \boldsymbol{y}) = k_t(\boldsymbol{y}, \boldsymbol{x})$
- Satisfy semigroup identity:

$$k_{t+s}(\boldsymbol{x}, \boldsymbol{y}) = \int_{\mathcal{M}} k_t(\boldsymbol{x}, \boldsymbol{z}) k_s(\boldsymbol{y}, \boldsymbol{z}) \mathrm{d}\boldsymbol{z}$$

- Isometric invariant
- Informative
- Multi-Scale
- Stable

Theorem (Intrinsic Property)

If $\boldsymbol{T} : \mathcal{M} \to \mathcal{N}$ is an isometry between two Riemannian manifolds $\mathcal{M}$ and $\mathcal{N}$, then

$$k_t^{\mathcal{M}}(\boldsymbol{x}, \boldsymbol{y}) = k_t^{\mathcal{N}}(\boldsymbol{T}(\boldsymbol{x}), \boldsymbol{T}(\boldsymbol{y}))$$

for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{M}$ and any $t > 0$.

Theorem (Informative Property[2])

Let $\boldsymbol{T} : \mathcal{M} \to \mathcal{N}$ be a surjective map between two Riemannian manifolds. If $k_t^{\mathcal{M}}(\boldsymbol{x}, \boldsymbol{y}) = k_t^{\mathcal{N}}(\boldsymbol{T}(\boldsymbol{x}), \boldsymbol{T}(\boldsymbol{y}))$ for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{M}$ and any $t > 0$, then $\boldsymbol{T}$ is an isometry.

Remark:

Follows from Varadhan's formula:

$$\lim_{t \to 0} t \log k_t(\boldsymbol{x}, \boldsymbol{y}) = -\frac{1}{4} d^2(\boldsymbol{x}, \boldsymbol{y}),$$

where d is the geodesic distance.

[2] A. Grigor'yan. "Heat kernels on weighted manifolds and applications". In: *Cont. Math* 398.2006 (2006), pp. 93–191.

Theorem (Multi-Scale Property[3])

- *For any smooth and relatively compact domain* $\mathcal{D} \subseteq \mathcal{M}$

$$\lim_{t \to 0} k_t^{\mathcal{D}}(\boldsymbol{x}, \boldsymbol{y}) = k_t(\boldsymbol{x}, \boldsymbol{y})$$

- *For any* $t \in \mathbb{R}^+$ *and any* $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{D}_1$*, the Dirichlet heat kernel*

$$k_t^{\mathcal{D}_1}(\boldsymbol{x}, \boldsymbol{y}) \leq k_t^{\mathcal{D}_2}(\boldsymbol{x}, \boldsymbol{y})$$

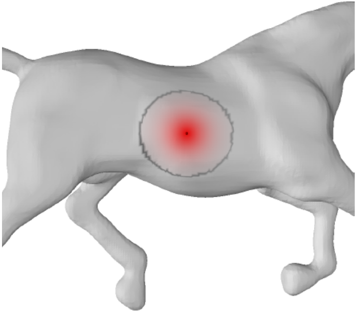
  if $\mathcal{D}_1 \subseteq \mathcal{D}_2$

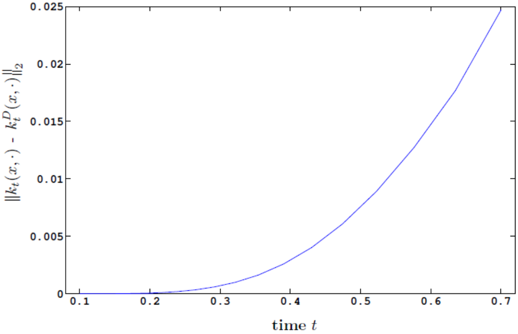
- *Moreover, if* $\{\mathcal{D}_n\}$ *is an expanding and exhausting sequence, i.e.,* $\mathcal{D}_{n-1} \subset \mathcal{D}_n$ *and* $\bigcup_{n=1}^{\infty} \mathcal{D}_n = \mathcal{M}$*, then*

$$\lim_{n \to \infty} k_t^{\mathcal{D}_n}(\boldsymbol{x}, \boldsymbol{y}) = k_t(\boldsymbol{x}, \boldsymbol{y})$$

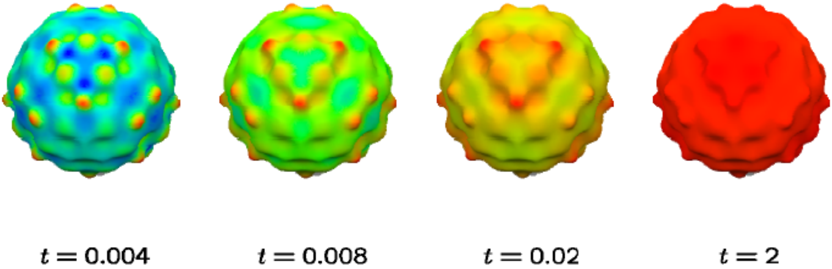
  for any t*.*

[3] A. Grigor'yan. "Heat kernels on weighted manifolds and applications". In: *Cont. Math* 398.2006 (2006), pp. 93–191.




0.025
0.02
0.015
0.01
0.005
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
time t
$\|k_t(x,\cdot) - k_t^D(x,\cdot)\|_2$

Can be interpreted as multi-scale intrinsic curvature[4]

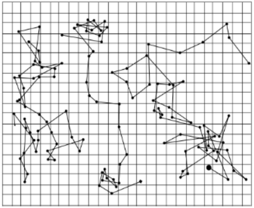
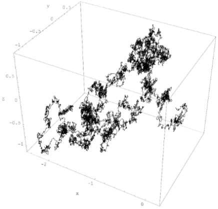
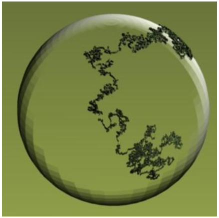


[4] J. Sun et al. "A concise and provably informative multi-scale signature based on heat diffusion". In: *Computer graphics forum*. Vol. 28. 5. Wiley Online Library. 2009, pp. 1383–1392.

- $k_t(\boldsymbol{x},\boldsymbol{y})$: probability density function of Brownian motion on the surface $\mathcal{M}$

$$\mathbb{P}\Big(W_x^t \in \mathcal{C}\Big) = \int_{\mathcal{C}} k_t(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y}$$

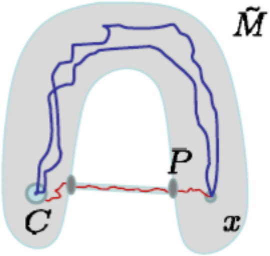
- Intuitively: weighted average over all possible paths between $\boldsymbol{x}$ and $\boldsymbol{y}$ in time t.

Z.-Q. Chen et al. "Stability and approximations of symmetric diffusion semigroups and kernels". In: *journal of functional analysis* 152.1 (1998), pp. 255–280

Robust:

- $k_t(\boldsymbol{x}, \cdot)$ is a probability density function, a weighted average over all paths, which should not be sensitive to local perturbations



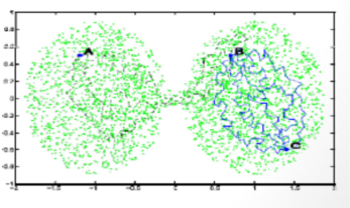
$$k_t^{\tilde{M}}(x, C) = \mathbb{P}(\tilde{W}_x^t \in C)$$

Z.-Q. Chen et al. "Stability and approximations of symmetric diffusion semigroups and kernels". In: *journal of functional analysis* 152.1 (1998), pp. 255–280

Diffusion distance[a]:

$$d_t^2(\boldsymbol{x},\boldsymbol{y}) = k_t(\boldsymbol{x},\boldsymbol{y}) + k_t(\boldsymbol{y},\boldsymbol{y}) - 2k_t(\boldsymbol{x},\boldsymbol{y})$$
$$= \sum_i \mathrm{e}^{-\lambda_i t}(\phi_i(\boldsymbol{x}) - \phi_i(\boldsymbol{y}))^2$$

[a] S. S. Lafon. *Diffusion maps and geometric harmonics*. Yale University, 2004.



S. Lafon

Eccentricity:

$$\mathrm{ecc}_t(\boldsymbol{x}) = \frac{1}{A_{\mathcal{M}}}\int_{\mathcal{M}} d_t^2(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y} = k_t(\boldsymbol{x},\boldsymbol{x}) + \sum_i \mathrm{e}^{-\lambda_i t} - \frac{2}{A_{\mathcal{M}}}$$

- $A_{\mathcal{M}}$ is the surface area.
- $\mathrm{ecc}_t(\boldsymbol{x})$ and $k_t(\boldsymbol{x},\boldsymbol{x})$ have the same level sets, in particular, extrema points.

- A single point signature $\{k_t(\boldsymbol{x}, \cdot)\}_{t \geq 0}$ is a function on $\mathbb{R}^+ \times \mathcal{M}$.
- Difficult to compare $\{k_t(\boldsymbol{x}_1, \cdot)\}_{t \geq 0}$ to $\{k_t(\boldsymbol{x}_2, \cdot)\}_{t \geq 0}$. The main problem is the unstable ordering of the Eigenfunctions of the Laplace-Beltrami Operator

Definition (Heat Kernel Signature (HKS))

The Heat Kernel Signature is a restriction of $\{k_t(\boldsymbol{x}, \cdot)\}_{t \geq 0}$ to the temporal domain:

$$\mathrm{HKS}_{\boldsymbol{x}}(t) : \mathbb{R}^+ \to \mathbb{R}^+$$

$$\mathrm{HKS}_{\boldsymbol{x}}(t) = k_t(\boldsymbol{x}, \boldsymbol{x}) = \sum_{i=1}^{\infty} \mathrm{e}^{-\lambda_i t} \phi_i(\boldsymbol{x}) \phi_i(\boldsymbol{x}) = \sum_{i=1}^{\infty} \mathrm{e}^{-\lambda_i t} \phi_i(\boldsymbol{x})^2$$

and $\lambda(t) = \mathrm{e}^{\lambda_i t}$.

- Concise and commensurable
- Multi-Scale
- Isometric invariant
- Stable (perturbations and meshing)
- Informative?

Theorem (Informative Property)

If the eigenvalues of the Laplace-Beltrami operators of two compact manifolds $\mathcal{M}$ and $\mathcal{N}$ are not repeated, and $\boldsymbol{T}$ is a homeomorphism from $\mathcal{M}$ to $\mathcal{N}$, then $\boldsymbol{T}$ is isometric if and only if

$$k_t^{\mathcal{M}}(\boldsymbol{x},\boldsymbol{x}) = k_t^{\mathcal{N}}(\boldsymbol{T}(\boldsymbol{x}),\boldsymbol{T}(\boldsymbol{x}))$$

for any $\boldsymbol{x} \in \mathcal{M}$ and any $t > 0$.

- Most shapes have no repeated eigenvalues.
- The theorem fails on a sphere or a torus.

Discrete case:

- Build the discrete Laplace operator $\boldsymbol{L} = \boldsymbol{A}^{-1}\boldsymbol{W}$
- Solve

$$\boldsymbol{W}\phi = \lambda \boldsymbol{A}\phi$$

- Compute

$$\mathrm{HKS}_{\boldsymbol{x}}(t) = \sum_i \mathrm{e}^{-\lambda_i t}\phi_i^2(\boldsymbol{x})$$

The polynomial expansion of $\mathrm{HKS}_{\boldsymbol{x}}(t)$ at small t:

$$\mathrm{HKS}_{\boldsymbol{x}}(t) = (4\pi t)^{-d/2}\left(1 + \frac{1}{6}s(\boldsymbol{x})t + \mathcal{O}\left(t^2\right)\right)$$

$s(\boldsymbol{x})$ is the Gaussian curvature on a surface

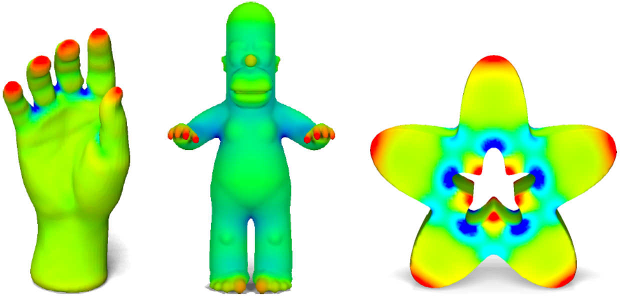


Figure: HKS for small fixed t

S. Minakshisundaram and Å. Pleijel. "Some properties of the eigenfunctions of the Laplace-operator on Riemannian manifolds". In: *Canadian Journal of Mathematics* 1.3 (1949), pp. 242–256

Rustamov et al., 2007

$$\mathrm{GPS}(\boldsymbol{p}) = \left(\frac{1}{\sqrt{\lambda_1}}\phi_1(\boldsymbol{p}), \frac{1}{\sqrt{\lambda_2}}\phi_2(\boldsymbol{p}), \frac{1}{\sqrt{\lambda_3}}\phi_3(\boldsymbol{p}), \ldots \right)$$

- Global vs. multi-scale
- Stability

$$\mathrm{HKS}_{\boldsymbol{x}}(t) = \sum_i \mathrm{e}^{-\lambda_i t} \phi_i^2(\boldsymbol{x})$$

How to compare descriptors at different locations?

- $\mathrm{HKS}_{\boldsymbol{x}}(t)$ decays exponentially as t increases.
- Variation of $\mathrm{HSK}_{\boldsymbol{x}}(t)$ is large for small t.

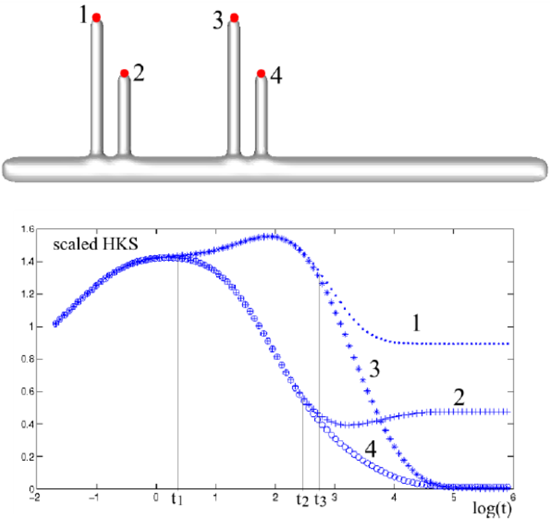
Solution:

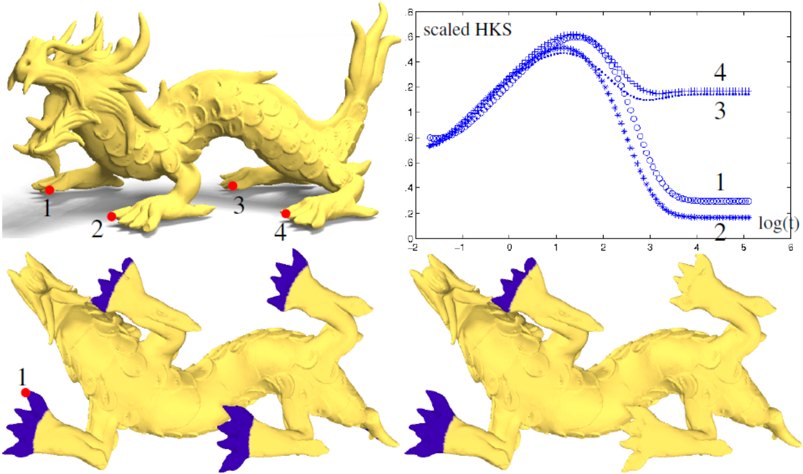
$$d_{[t_1,t_2]}(\boldsymbol{x}_1,\boldsymbol{x}_2) = \left(\int_{t_1}^{t_2} \left(\frac{|k_t(\boldsymbol{x}_1,\boldsymbol{x}_1) - k_t(\boldsymbol{x}_2,\boldsymbol{x}_2)|}{\int_{\mathcal{M}} k_t(\boldsymbol{x},\boldsymbol{x})\mathrm{d}\boldsymbol{x}} \right)^2 \mathrm{d}\log t \right)^{1/2}$$

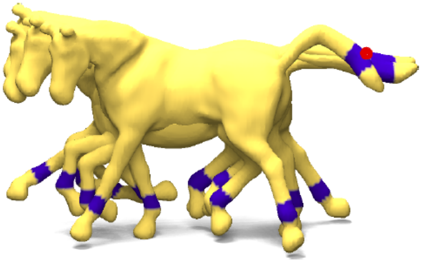
- Note that

$$\int_{\mathcal{M}} k_t(\boldsymbol{x},\boldsymbol{x})\mathrm{d}\boldsymbol{x} = \sum_{i=1}^{\infty} \mathrm{e}^{-\lambda_i}$$

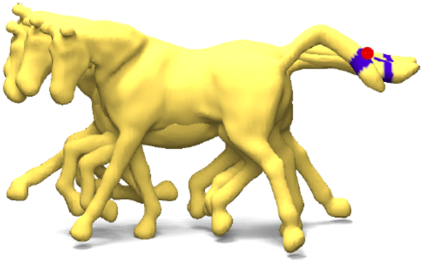
J. Sun et al. "A concise and provably informative multi-scale signature based on heat diffusion". In: *Computer graphics forum*. Vol. 28. 5. Wiley Online Library. 2009, pp. 1383–1392


1
2
3
4
scaled HKS
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-2
-1
0
t_1
1
2
t_2
t_3
3
4
5
6
log(t)
1
3
2
4


scaled HKS
4
3
1
2
log(t)
−2 −1 0 1 2 3 4 5 6
1
2
3
4
1

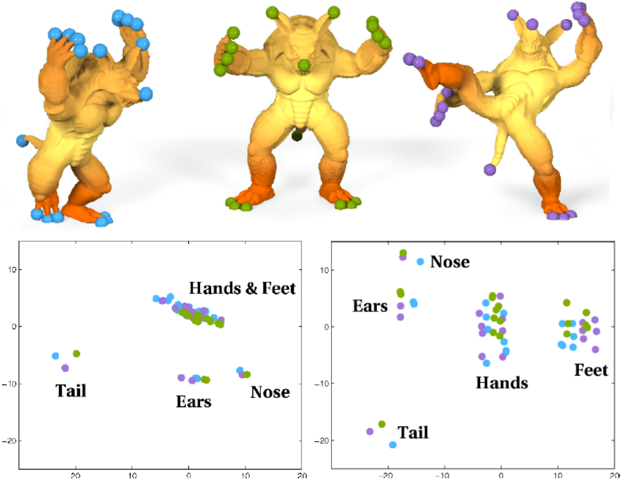


Half of ts



All ts

A local maxima of $k_t(x, x)$ for a large t.



$t = [0.1,\ 4]$ $\quad$ $t = [0.1,\ 80]$

- There are more spectral shape descriptors which are constructed by taking the diagonal values of heat-like operators.
- A generic descriptor of this kind has the form

$$f(\boldsymbol{x}) = \sum_{k=1}^{\infty} \boldsymbol{\tau}(\lambda_k)\phi_k^2(\boldsymbol{x}) \approx \sum_{k=1}^{K} \boldsymbol{\tau}(\lambda_k)\phi_k^2(\boldsymbol{x})$$

where

$$\boldsymbol{\tau}(\lambda) = (\tau_1(\lambda), \ldots, \tau_Q(\lambda))$$

is a bank of transfer functions acting on LBO eigenvalues, and Q is the descriptor dimensionality.

- Idea: Evaluate the probability of a quantum particle with a certain energy distribution to be located at a point $\boldsymbol{x}$.
- The behavior of a quantum particle on a surface is governed by the Schrödinger equation
$$\left(\mathrm{i}\Delta_{\mathcal{M}} + \frac{\partial}{\partial t}\right)\Psi(\boldsymbol{x}, t) = 0$$
where $\Psi(\boldsymbol{x}, t)$ is the complex wave function.
- Note that instead of representing a diffusion, the solution now has oscillatory behavior.

M. Aubry et al. "The wave kernel signature: A quantum mechanical approach to shape analysis". In: *2011 IEEE international conference on computer vision workshops (ICCV workshops)*. IEEE. 2011, pp. 1626–1633

- Let us assume that the quantum particle has an initial energy distribution $f(e) \approx f(\nu)$ where ν is the frequency.
- The solution to the Schrödinger equation can then be expressed in the spectral domain of $\Delta_{\mathcal{M}}$ as

$$\Psi(\boldsymbol{x}, t) = \sum_{k \geq 1} \exp(\mathrm{i}\nu_k t) f(\nu_k) \phi_k(\boldsymbol{x})$$

- The probability to measure the particle at a point $\boldsymbol{x}$ at time t is given by

$$|\Psi(\boldsymbol{x}, t)|^2$$

- By integrating over all times, the average probability to measure the particle at a point $\boldsymbol{x}$ is obtained:

$$p(\boldsymbol{x}) = \lim_{T \to \infty} \frac{1}{T} \int_0^T |\Psi(\boldsymbol{x}, t)|^2 \mathrm{d}t = \sum_{k \geq 1} f^2(\nu_k) \phi_k^2(\boldsymbol{x})$$

Note that the probability distribution depends on the initial energy distribution f.

- For the WKS, a family of log-normal energy distributions centered around some mean log energy $\log e$ with variance σ is considered.

$$f_e(\nu) = \exp\left(-\frac{\log e - \log \nu}{2\sigma^2}\right)$$

- This particular choice of distributions is motivated by a perturbation analysis for the Laplacian spectrum.
- Fixing the family of energy distributions, each point on the surface is associated with a wave kernel signature of the form

$$p(\boldsymbol{x}) = \left(p_{e_1}(\boldsymbol{x}), \ldots, p_{e_n}(\boldsymbol{x})\right)^\top$$

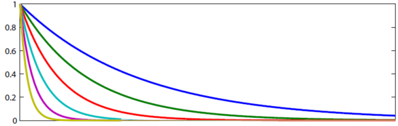
where $p(\boldsymbol{x})$ is the probability to measure a quantum particle with the initial energy distribution $f_e(\nu)$ at point $\boldsymbol{x}$.

Wave kernel signatures

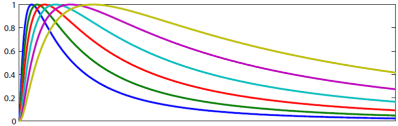
- The WKS descriptor resembles the HKS descriptor in the sense that it can also be thought of as an application of a set of filters with the frequency responses $f_e^2(\nu)$.

$$\mathrm{HKS}_{\boldsymbol{x}}(t) = \sum_{k=1}^{\infty} \mathrm{e}^{-\lambda_k t} \phi_k(\boldsymbol{x})^2 \quad p_e(\boldsymbol{x}) = \sum_{k=1}^{\infty} f_e^2(\nu_k) \phi_k^2(\boldsymbol{x})$$

- However, unlike the HKS that uses low-pass filters, the responses of the WKS are band-pass.

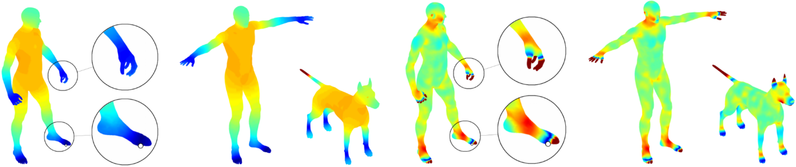


(a) Examples of (unnormalized) kernels used for the computation of the heat kernel.



(b) Examples of (unnorimalized) kernels used for the computation of the wave kernel.

- The band pass filter reduces the influence of the low frequencies and allows better separation of frequency bands across the descriptor dimensions.
- As the result, the wave kernel descriptor exhibits superior feature localisation but at the same time tend to produce noisier matches.



(a) HKS (b) WKS

Normalized Euclidean distance between the descriptor at a reference point on the left foot (white dot in the leftmost column) and descriptors computed at rest of the points of the same shape (left column), its approximate isometry (middle column), and a distinct shape (right column).

Optimal spectral descriptors

Based on the general representation

$$f(\boldsymbol{x}) = \sum_{k=1}^{\infty} \tau(\lambda_k)\phi_k^2(\boldsymbol{x}) \approx \sum_{k=1}^{K} \tau(\lambda_k)\phi_k^2(\boldsymbol{x}),$$

OSD use parametric transfer functions expressed as

$$\tau_q(\lambda) = \sum_{i=1}^{M} \alpha_{qm}\beta_m(\lambda)$$

in the B-spline basis $\beta_1(\lambda), \ldots, \beta_M(\lambda)$, where

$$\alpha_{qm}(q = 1, \ldots, Q, m = 1, \ldots, M)$$

are the parameterization coefficients.

R. Litman and A. M. Bronstein. "Learning spectral descriptors for deformable shape correspondence". In: *IEEE transactions on pattern analysis and machine intelligence* 36.1 (2013), pp. 171–180

- Plugging the coefficients into the general representation we get

$$f_q(\boldsymbol{x}) = \sum_{k=1}^{\infty} \tau_q(\lambda_k)\phi_k^2(\boldsymbol{x}) = \sum_{m=1}^{M} \alpha_{qm} \underbrace{\sum_{k=1}^{K} \beta_m(\lambda_k)\phi_k^2(\boldsymbol{x})}_{g_m(\boldsymbol{x})}$$

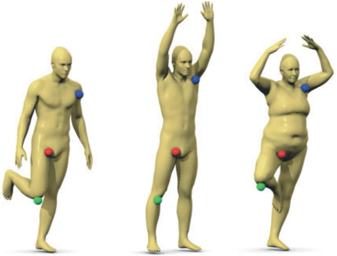
- The vector $\boldsymbol{g}(\boldsymbol{x}) = (g_1(\boldsymbol{x}), \ldots, g_M(\boldsymbol{x}))^\top$ is called geometry vector. It depends only on the intrinsic geometry of the shape.
- Writing the coefficients as $Q \times M$-Matrix $\boldsymbol{A} = (\alpha_{qm})$ leads to

$$\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{f}_{\boldsymbol{A}}(\boldsymbol{x})\boldsymbol{A}\boldsymbol{g}(\boldsymbol{x})$$

- The main idea of Litman and Bronstein is to learn the optimal parameters $\boldsymbol{A}$ by minimizing task-specific loss which reduces to a Mahalanobis-type metric learning.

X. He et al. "Learning a maximum margin subspace for image retrieval". In: *IEEE Transactions on Knowledge and Data Engineering* 20.2 (2007), pp. 189–201

- Given a set of points $\boldsymbol{x}$, knowingly similar points (positive) $\boldsymbol{x}_+$ and knowingly dissimilar points (negative) $\boldsymbol{x}_-$, with respective geometry vectors $\boldsymbol{g}, \boldsymbol{g}_+, \boldsymbol{g}_-$



$$\min_{\boldsymbol{A}, \boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}} \alpha \mathbb{E}\left[\|\boldsymbol{f}_{\boldsymbol{A}}(\boldsymbol{x}) - \boldsymbol{f}_{\boldsymbol{A}}(\boldsymbol{x}_+)\|^2 \right] - (1-\alpha) \mathbb{E}\left[\|\boldsymbol{f}_{\boldsymbol{A}}(\boldsymbol{x}) - \boldsymbol{f}_{\boldsymbol{A}}(\boldsymbol{x}_-)\|^2 \right]$$

- Find optimal $\boldsymbol{A}$ by minimizing the loss.

M. Defferrard et al. "Convolutional neural networks on graphs with fast localized spectral filtering". In: *Advances in neural information processing systems* 29 (2016)

- Optimal transfer functions

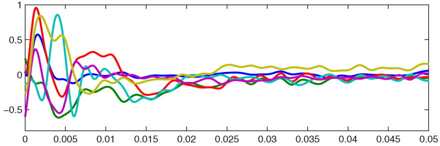
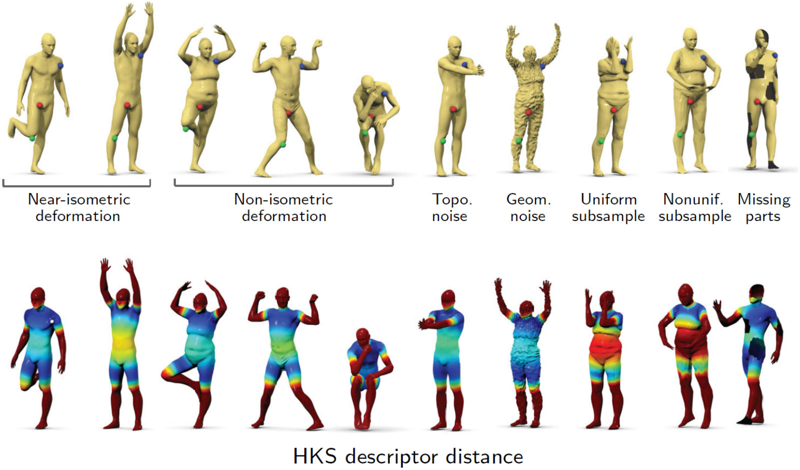


Figure: Optimal transfer function learned from positive and negative examples

Figure: Optimal transfer function learned from positive and negative examples

From: Deep Learning for Shape Analysis, EUROGRAPHICS Tutorial, Lisbon, May 2016, `http://geometricdeeplearning.com/`

- Descriptor robustness



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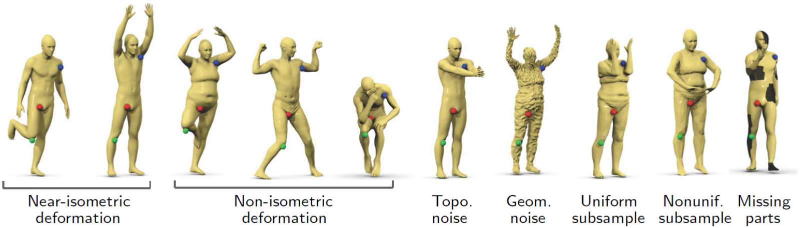
- Descriptor robustness



WKS descriptor distance

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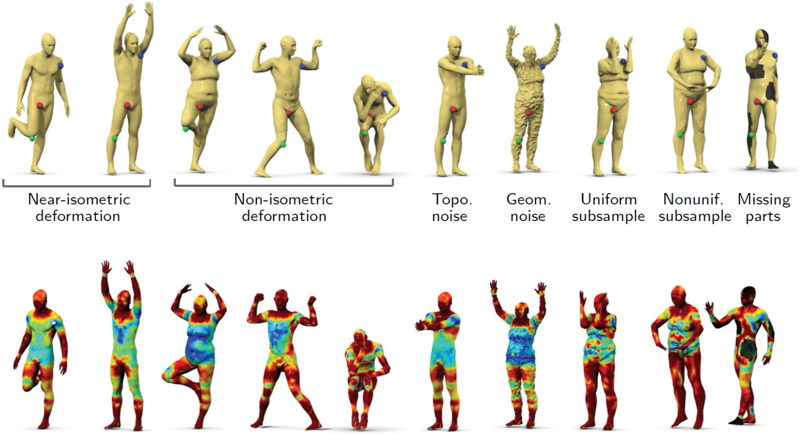
- Descriptor robustness



Optimal Spectral descriptor distance

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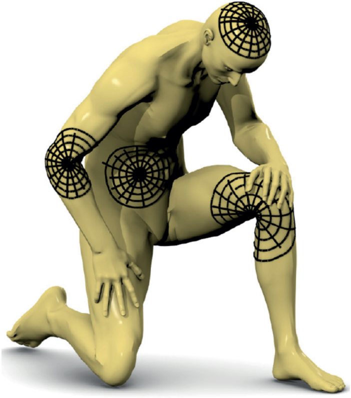
Geodesic polar coordinates

- Local system of geodesic polar coordinates at $\boldsymbol{x}$
- ρ-level set of geodesic distance function $d_X(\boldsymbol{x}, \xi)$ truncated at ρ_0
- Points along geodesic $\Gamma_\theta(\boldsymbol{x})$ emanating from $\boldsymbol{x}$ in direction θ
- Local chard: bijective map $\Omega(\boldsymbol{x}) : B_{\rho_0}(\boldsymbol{x}) \to [0, \rho_0] \times [0, 2\pi)$ from manifold to local coordinates (ρ, θ) around $\boldsymbol{x}$
- Patch operator applied to $f \in L^2(X)$

$$(D(\boldsymbol{x})f)(\rho, \theta) = (f \circ \Omega^{-1}(\boldsymbol{x}))(\rho, \theta)$$



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Patch operator construction

$$(D\boldsymbol{x}f)(\rho,\theta) = \frac{\int_X \nu_\rho(\boldsymbol{x},\xi)\nu_\theta(\boldsymbol{x},\xi)f(\xi)}{\int_X \nu_\rho(\boldsymbol{x},\xi)\nu_\theta(\boldsymbol{x},\xi)\mathrm{d}\xi}$$

Radial weight

$$\nu_\rho(\boldsymbol{x},\xi) \propto \mathrm{e}^{-(d_X(\boldsymbol{x},\xi)-\rho)^2/\sigma_\rho^2}$$

Angular weight

$$\nu_\theta(\boldsymbol{x},\xi) \propto \mathrm{e}^{-d_X^2(\Gamma(\boldsymbol{x},\theta),\xi)/\sigma_\theta^2}$$

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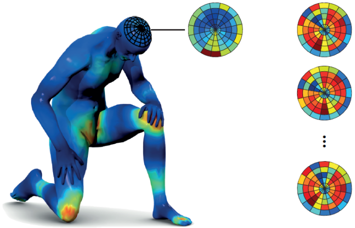
- Geodesic convolution: apply filter a to patches extracted from $f \in L^2(X)$ in local geodesic polar coordinates

$$(f * a) = \sum_{\theta,r} \underbrace{(D(\boldsymbol{x})f)(r,\theta)}\, \underbrace{\alpha(\theta + \Delta\theta, r)}$$



- Angular coordinate arbitrary: rotation ambiguity!
- Keep all possible rotations

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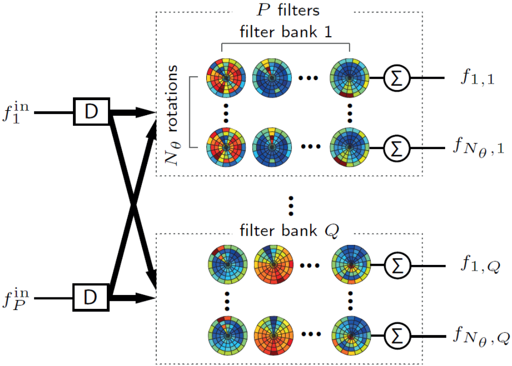

$$f^{\text{out}}_{\Delta\theta,q}(\theta,r) = \sum_{p=1,\dots,P} (f * a_{\Delta\theta,qp})(\boldsymbol{x}), \quad q = 1,\dots,Q$$

$a_{\Delta\theta,qp} = a_{qp}(\theta + \Delta\theta, r)$ are the coefficients of the p-th filter in q-th filterbank rotated by $\Delta\theta$

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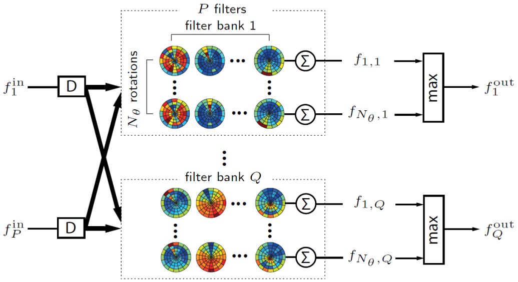

$$f^{\text{out}}_{\Delta\theta,q}(\theta, r) = \sum_{p=1,\ldots,P} (f * a_{\Delta\theta,qp})(\boldsymbol{x}), \quad q = 1, \ldots, Q$$

$a_{\Delta\theta,qp} = a_{qp}(\theta + \Delta\theta, r)$ are the coefficients of the p-th filter in q-th filterbank rotated by $\Delta\theta$

Use angular max pooling to remove rotation ambiguity!

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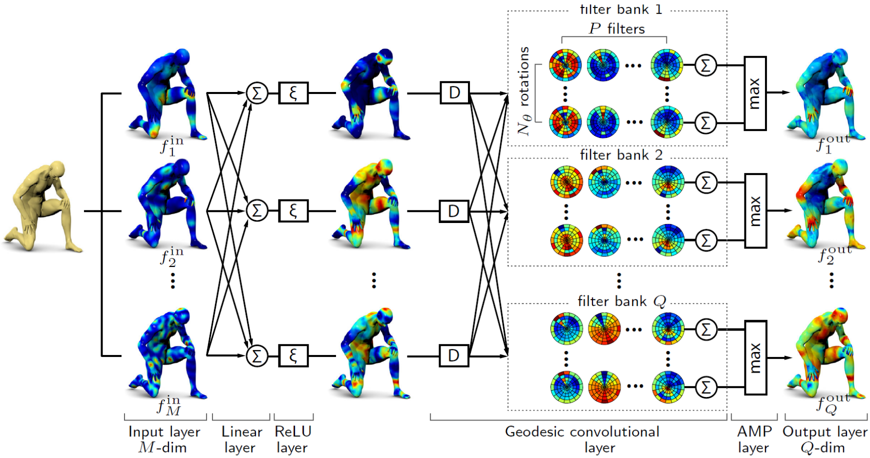

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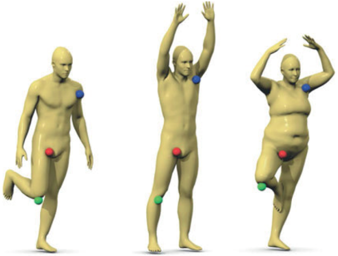
- Given a set of points $\boldsymbol{x}$, knowingly similar points (positive) $\boldsymbol{x}_+$ and knowingly dissimilar points (negative) $\boldsymbol{x}_-$, with respective geometry vectors $\boldsymbol{g}, \boldsymbol{g}_+, \boldsymbol{g}_-$

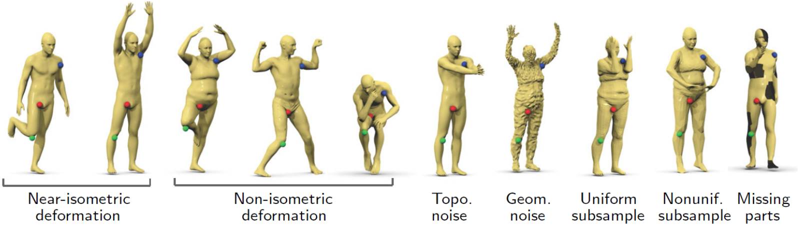


- Minimize siamese loss w.r.t. GCNN parameters Θ

$$l(\Theta) = (1-\alpha) \sum_{(\boldsymbol{x},\boldsymbol{x}_+)\in X^+} [\|\boldsymbol{f}_\Theta(\boldsymbol{x}) - \boldsymbol{f}_\Theta(\boldsymbol{x}_+)\|] - \alpha \sum_{(\boldsymbol{x},\boldsymbol{x}_-)\in X^-} [\|\boldsymbol{f}_\Theta(\boldsymbol{x}) - \boldsymbol{f}_\Theta(\boldsymbol{x}_-)\|]$$

J. Masci et al. "Geodesic convolutional neural networks on riemannian manifolds". In: *Proceedings of the IEEE international conference on computer vision workshops*. 2015, pp. 37–45

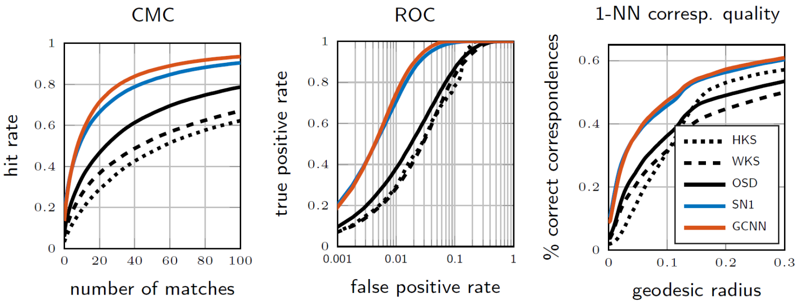
- Descriptor robustness



GCNN descriptor distance

From: Deep Learning for Shape Analysis, EUROGRAPHICS Tutorial, Lisbon, May 2016, `http://geometricdeeplearning.com/`

Descriptor Performance

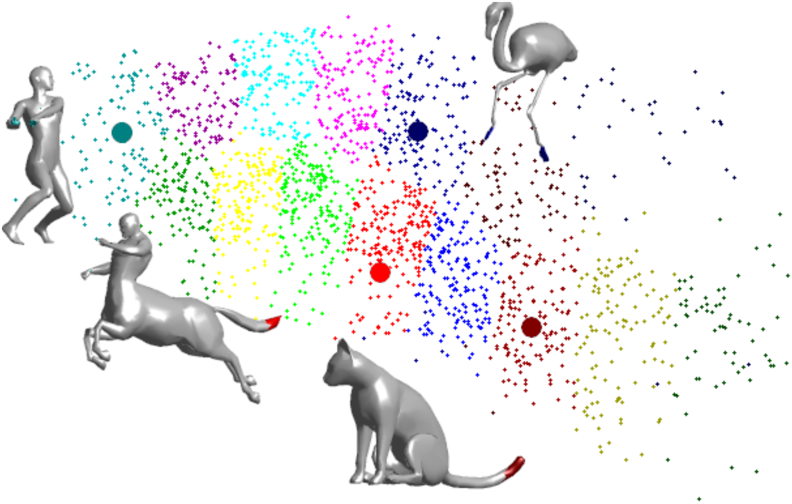


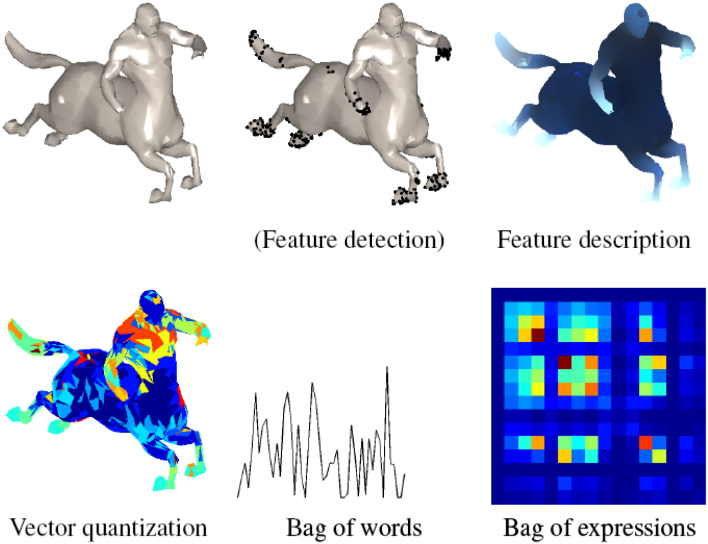
Descriptor performance using symmetric Princeton benchmark
(training and testing: FAUST)

Cumulative match characteristic (CMC) evaluates the probability of a correct corre-spondence among theknearest neighbors in the descriptorspace. Receiver operator characteristic (ROC) measures the percentage of positives andnegatives pairs falling below various thresholds of their dis-tance in the descriptor space (true positiveandnegativerates, respectively).

J. Masci et al. "Geodesic convolutional neural networks on riemannian manifolds". In: *Proceedings of the IEEE international conference on computer vision workshops*. 2015, pp. 37–45

Geometric Vocabulary




(Feature detection)
Feature description
Vector quantization
Bag of words
Bag of expressions

Vocabulary of representative HKS

- Compute normalized multi-scale HKS $\boldsymbol{P}(\boldsymbol{x})$ for points $\boldsymbol{x}$ on the objects:

$$\boldsymbol{P}(\boldsymbol{x}) = (p_1(\boldsymbol{x}), \dots, p_n(\boldsymbol{x}))^\top$$
$$p_i(\boldsymbol{x}) = c(x) k_{\alpha^{i-1} t_0}(\boldsymbol{x}, \boldsymbol{x})$$

with $c(x)$ constant such that $\|\boldsymbol{P}(\boldsymbol{x})\|_2 = 1$

- Cluster the HKS (k-means) into k cluster
- Build vocabulary

$$\mathcal{V} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$$

A. M. Bronstein et al. “Shape google: Geometric words and expressions for invariant shape retrieval”. In: *ACM Transactions on Graphics (TOG)* 30.1 (2011), pp. 1–20

M. Ovsjanikov, A. M. Bronstein, et al. “Shape google: a computer vision approach to isometry invariant shape retrieval”. In: *2009 IEEE 12th International Conference on Computer Vision Workshops, ICCV Workshops*. IEEE. 2009, pp. 320–327

Given a geometric vocabulary $\mathcal{P} = \{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_V\}$ for each point with local descriptor $\boldsymbol{p}(\boldsymbol{x})$ find **representation** $\theta(\boldsymbol{x}) = (\theta_1(\boldsymbol{x}), \ldots, \theta_V(\boldsymbol{x}))^\top$

- **Hard quantization**

$$k^* = \underset{i=1,\ldots,V}{\arg\min} \|\boldsymbol{p}(\boldsymbol{x}) - \boldsymbol{p}_i\|$$

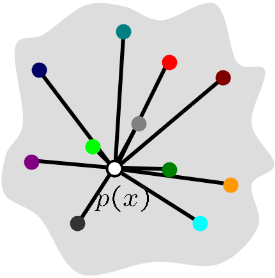
$$\theta_k(\boldsymbol{x}) = \begin{cases} 1, & k = k^* \\ 0, & \text{else} \end{cases}$$

Nearest neighbor in the descriptor space

- **Soft quantization**

$$\theta_k(\boldsymbol{x}) = \mathrm{e}^{-\frac{\|\boldsymbol{p}(\boldsymbol{x}) - \boldsymbol{p}_k\|_2^2}{2\sigma^2}}$$

Weighted distance to words in vocabulary

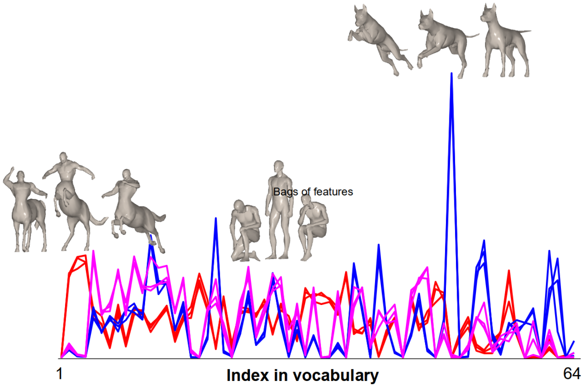


Statistics of different geometric words over the entire shape by integration

$$f(X) = \int_X \theta(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}$$

Shape distance = distance between bags of features

$$d_{\mathrm{BoF}}(X, Y) = \|f(X) - f(Y)\|$$



- Compare objects using their BoFs:

$$d_{\mathrm{BoF}}(X, Y) = \|\mathrm{BoF}(X) - \mathrm{BoF}(Y)\|_1$$

- Spatially sensitive BoF (considering word pairs)

$$\mathrm{SSBoF}(X) = \int_{\mathcal{M}} \int_{\mathcal{M}} k_t(\boldsymbol{x}, \boldsymbol{y}) \Theta(\boldsymbol{x}) \Theta(\boldsymbol{y}) \mathrm{d}A \mathrm{d}A \approx \sum_{x, y \in \mathcal{M}} k_t(\boldsymbol{x}, \boldsymbol{y}) \Theta(\boldsymbol{x}) \Theta(\boldsymbol{y})$$

- Statistical weighting
  - Shape database of size D, shapes $X_1, \ldots, X_D$
  - Inverse document frequency

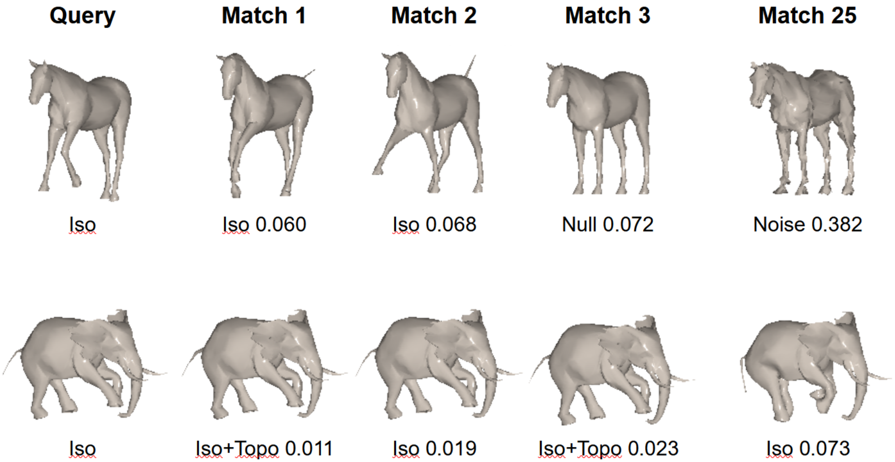
$$\omega_i = \log \left(\frac{D}{\sum_{j=1}^{D} \delta(f_i(X_j) > 0)} \right)$$

  where δ is an indicator function, and $f_i(X_j)$ counts the number of occurrences of word i in shape j

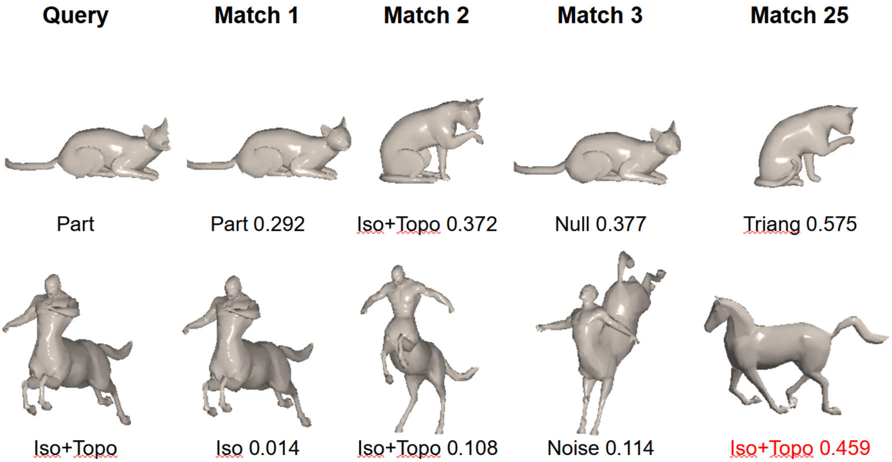
- Weighted L_1-Distance

$$d_{\mathrm{BoFw}}(X, Y) = \sum_{i=1}^{V} |f_i(X) - f_i(Y)|$$

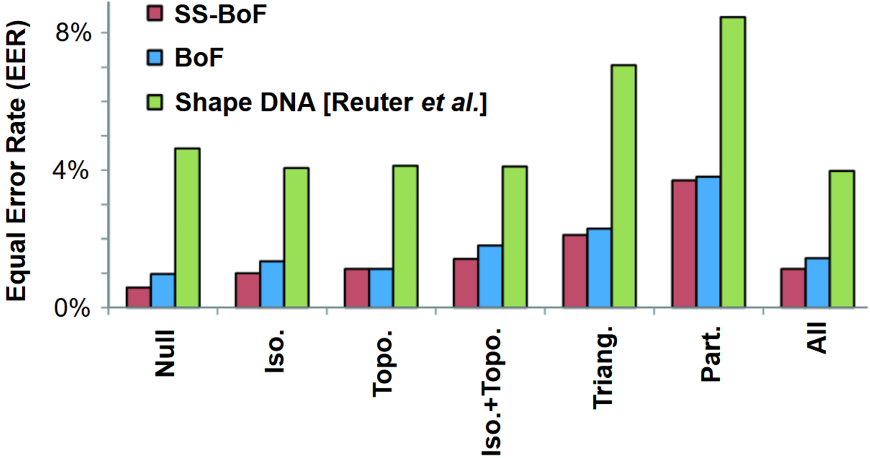
M. Ovsjanikovs. *Spectral methods for isometric shape matching and symmetry detection*. Stanford University, 2011


Query
Match 1
Match 2
Match 3
Match 25
Iso
Iso 0.060
Iso 0.068
Null 0.072
Noise 0.382
Iso
Iso+Topo 0.011
Iso 0.019
Iso+Topo 0.023
Iso 0.073

Shape google

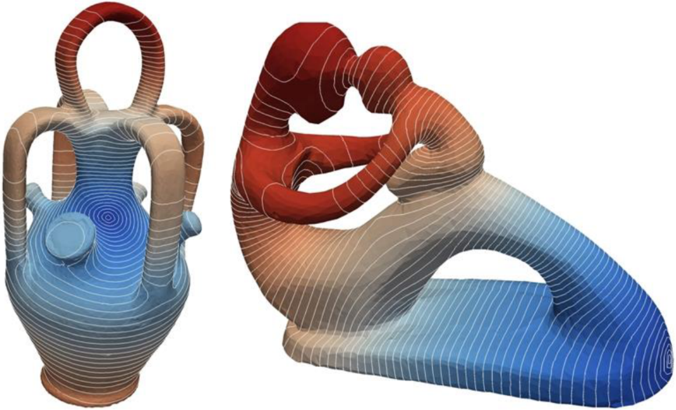


Comparison



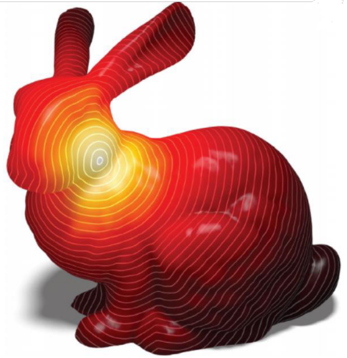
SS-BoF
BoF
Shape DNA [Reuter et al.]
Equal Error Rate (EER)
8%
4%
0%
Null
Iso.
Topo.
Iso.+Topo.
Triang.
Part.
All

$$d_b(\boldsymbol{x}, \boldsymbol{y}) := \|g_x - g_y\|_2, \text{ where } \Delta g_x = \delta_x$$



"Biharmonic distance"
Lipman, Rustamov & Funkhouser, 2010

$$d_g(\boldsymbol{x}, \boldsymbol{y}) := \lim_{t \to 0} \sqrt{-4t \log k_t(\boldsymbol{x}, \boldsymbol{y})}$$



"Geodesics in heat"
Crane, Weischedel, and Wardetzky; TOG 2013

$$d_g(\boldsymbol{x},\boldsymbol{y}) := \lim_{t\to 0} \sqrt{-4t \log k_t(\boldsymbol{x},\boldsymbol{y})}$$

Algorithm 1 The Heat Method

I. Integrate the heat flow $\dot{u} = \Delta u$ for time t.
II. Evaluate the vector field $X = -\nabla u / |\nabla u|$.
III. Solve the Poisson equation $\Delta\phi = \nabla \cdot X$.

Crane, Weischedel, and Wardetzky. "Geodesics in Heat." TOG, 2013.

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